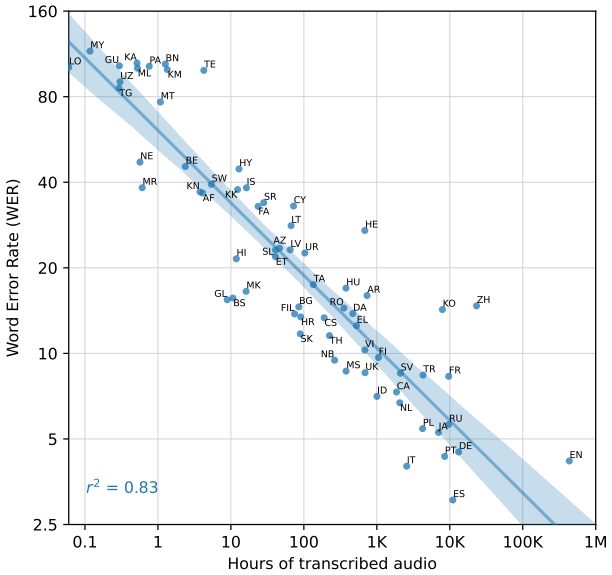


Figure 3. **Correlation of pre-training supervision amount with downstream speech recognition performance.** The amount of pre-training speech recognition data for a given language is very predictive of zero-shot performance on that language in Fleurs.

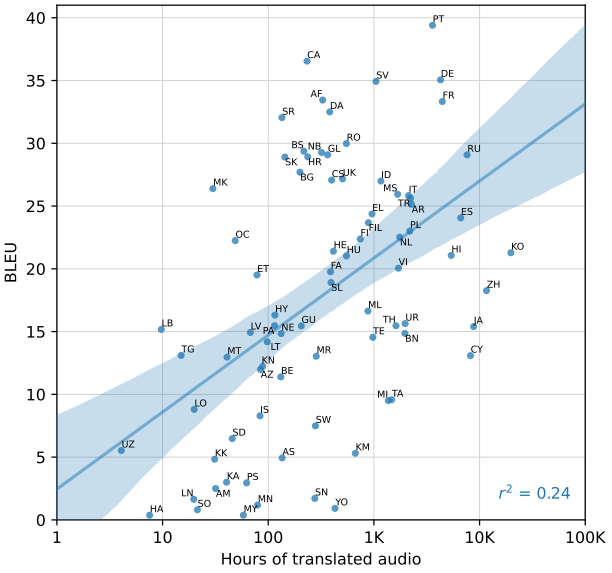


Figure 4. **Correlation of pre-training supervision amount with downstream translation performance.** The amount of pre-training translation data for a given language is only moderately predictive of Whisper's zero-shot performance on that language in Fleurs.

Model	MLS	VoxPopuli
VP-10K + FT	-	15.3
XLS-R (1B)	10.9	10.6
mSLAM-CTC (2B)	9.7	9.1
Maestro	-	**8.1**
Zero-Shot Whisper	**7.3**	13.6

Table 3. **Multilingual speech recognition performance.** Zero-shot Whisper improves performance on Multilingual LibriSpeech (MLS) but is still significantly behind both Maestro, XLS-R, and mSLAM on VoxPopuli.

et al., 2022), and Maestro (Chen et al., 2022b) in a zero-shot setting. We caution that we do use a simple text standardizer for this result which prevents direct comparison or claims of SOTA performance. On VoxPopuli, however, Whisper significantly underperforms prior work and only beats the VP-10K+FT baseline from the original paper. We suspect the underperformance of Whisper models on VoxPopuli could be due to other models including this distribution as a major source for their unsupervised pre-training data and the dataset having significantly more supervised data, which benefits fine-tuning. While MLS has 10 hours of training data per language, the average amount of training data per language is roughly 10× higher for VoxPopuli.

These two benchmarks are somewhat narrow since they only include 15 unique languages, almost all of which are in the Indo-European language family and many of which are high-resource languages. These benchmarks only provide limited coverage and room to study Whisper models multilingual capabilities which include training data for speech recognition in 75 languages. To study the performance of Whisper more broadly we also report performance on the Fleurs dataset (Conneau et al., 2022). In particular, we were interested in studying the relationship between the amount of training data we have for a given language and the resulting downstream zero-shot performance for that language. We visualize this relation in Figure 3. We find a strong squared correlation coefficient of 0.83 between the log of the word error rate and the log of the amount of training data per language. Checking the regression coefficient for a linear fit to these log-log values results in an estimate that WER halves for every 16× increase in training data. We also observed that many of the largest outliers in terms of worse than expected performance according to this trend are languages that have unique scripts and are more distantly related to the Indo-European languages making up the majority of the training dataset such as Hebrew (HE), Telugu (TE), Chinese (ZH), and Korean (KO). These differences could be due to a lack of transfer due to linguistic distance, our byte level BPE tokenizer being a poor match for these languages, or variations in data quality.

X → English	High	Mid	Low	All
XMEF-X	34.2	20.2	5.9	14.7
XLS-R (2B)	36.1	27.7	15.1	22.1
mSLAM-CTC (2B)	37.8	29.6	18.5	24.8
Maestro	**38.2**	31.3	18.4	25.2
Zero-Shot Whisper	36.2	**32.6**	**25.2**	**29.1**

Table 4. **X→en Speech translation performance.** Zero-shot Whisper outperforms existing models on CoVoST2 in the overall, medium, and low resource settings but still moderately underperforms on high-resource languages compared to prior directly supervised work.

Language ID	Fleurs
w2v-bert-51 (0.6B)	71.4
mSLAM-CTC (2B)	**77.7**
Zero-shot Whisper	64.5

Table 5. **Language identification performance.** Zero-shot Whisper's accuracy at language identification is not competitive with prior supervised results on Fleurs. This is partially due to Whisper being heavily penalized for having no training data for 20 of Fleurs languages.

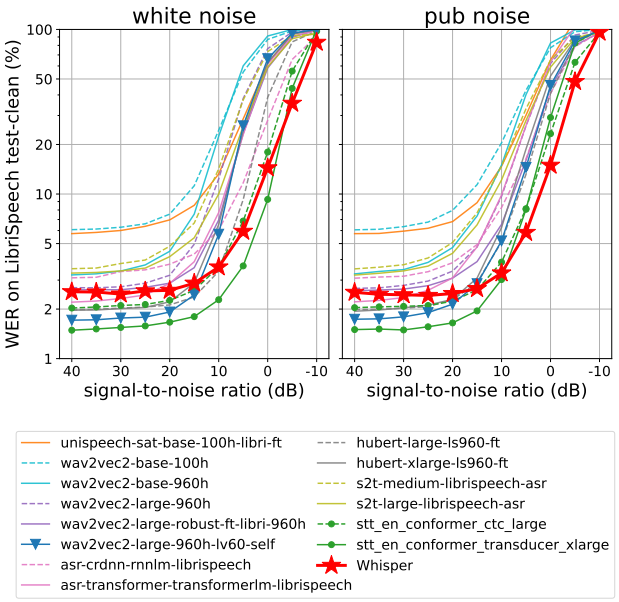


Figure 5. **WER on LibriSpeech test-clean as a function of SNR under additive white noise (left) and pub noise (right).** The accuracy of LibriSpeech-trained models degrade faster than the best Whisper model (★). NVIDIA STT models (●) perform best under low noise but are outperformed by Whisper under high noise (SNR < 10 dB). The second-best model under low noise (▼) is fine-tuned on LibriSpeech only and degrades even more quickly.

3.5. Translation

We study the translation capabilities of Whisper models by measuring their performance on the X→en subset of CoVoST2 (Wang et al., 2020b). We compare with Maestro, mSLAM, and XLS-R, the highest-performing prior work. We achieve a new state of the art of 29.1 BLEU zero-shot without using any of the CoVoST2 training data. We attribute this to the 68,000 hours of X→en translation data for these languages in our pre-training dataset which, although noisy, is vastly larger than the 861 hours of training data for X→en translation in CoVoST2. Since Whisper evaluation is zero-shot, it does particularly well on the lowest resource grouping of CoVoST2, improving over mSLAM by 6.7 BLEU. Conversely, the best Whisper model does not actually improve over Maestro and mSLAM on average for the highest resource languages.

For an additional analysis on an even wider set of languages, we also re-purpose Fleurs, which is a speech recognition dataset, as a translation dataset. Since the same sentences are transcribed for every language we use the English transcripts as reference translations. In Figure 4 we visualize the correlation between the amount of translation training data per language and the resulting zero-shot BLEU score on Fleurs. While there is a clear trend of improvement with increasing training data, the squared correlation coefficient is much lower than the 0.83 observed for speech recognition and only 0.24. We suspect this is partly caused by the noisier training data due to errors in audio language identification. As an example, Welsh (CY) is an outlier with much worse than expected performance at only 13 BLEU despite supposedly having 9,000 hours of translation data. This large amount of Welsh translation data is surprising, ranking 4th overall for translation data and ahead of some of the most spoken languages in the world like French, Spanish, and Russian. Inspection shows the majority of supposedly Welsh translation data is actually English audio with English captions where the English audio was mis-classified as Welsh by the language identification system, resulting in it being included as translation training data rather transcription data according to our dataset creation rules.

3.6. Language Identification

To evaluate language identification, we use the Fleurs dataset (Conneau et al., 2022). The zero-shot performance of Whisper is not competitive with prior supervised work here and underperforms the supervised SOTA by 13.6%. However, Whisper is heavily disadvantaged for language identification on Fleurs, since the Whisper dataset contains no training data for 20 of the 102 languages in Fleurs, upper-bounding accuracy at 80.4%. On the 82 overlapping languages the best Whisper model achieves 80.3% accuracy.

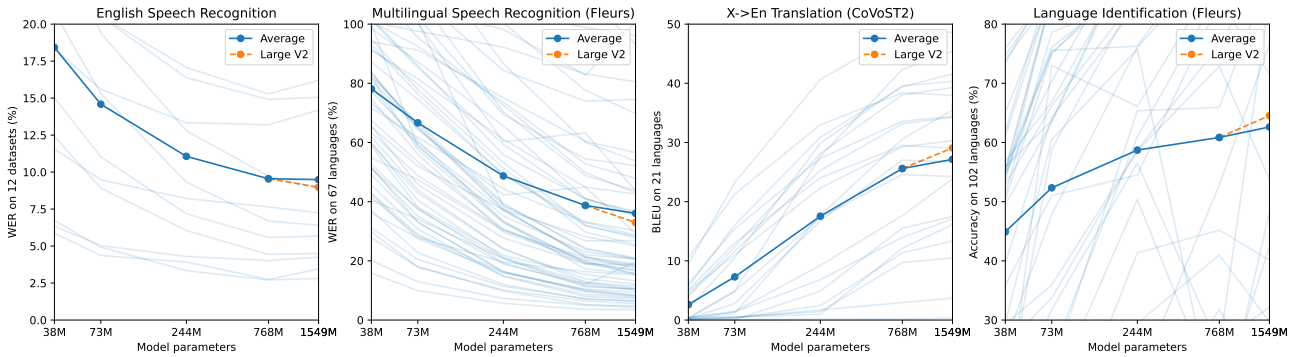


Figure 8. **Zero-shot Whisper performance scales reliably across tasks and languages with increasing model size.** Lightly shaded lines represent individual datasets or languages, showing that performance is more varied than the smooth trends in aggregate performance. Large V2 distinguished with a dashed orange line since it includes several changes that are not present for the smaller models in this analysis.

Dataset size	English WER (↓)	Multilingual WER (↓)	X→En BLEU (↑)
3405	30.5	92.4	0.2
6811	19.6	72.7	1.7
13621	14.4	56.6	7.9
27243	12.3	45.0	13.9
54486	10.9	36.4	19.2
681070	**9.9**	**29.2**	**24.8**

Table 6. **Performance improves with increasing dataset size.** English speech recognition performance refers to an average over 12 datasets while the Multilingual speech recognition reports performance on the overlapping subset of languages in Fleurs and X→en translation reports average BLEU on CoVoST2. Dataset size reported in hours.

All increases in the dataset size result in improved performance on all tasks, although we see significant variability in improvement rates across tasks and sizes. Performance improves rapidly on English speech recognition from 3,000 to 13,000 hours and then slows down noticeably between 13,000 and 54,000 hours. Using the full dataset, which corresponds to another 12.5× increase in size results in only a further 1 point drop in WER. This mirrors the diminishing returns observed with model size scaling for English speech recognition and could similarly be explained by saturation effects when approaching human-level performance.

Improvements in WER follow a power-law trend for multilingual speech recognition till 54,000 hours and then deviate from this trend, improving only a further 7 points when increasing to the full dataset size. For X→en translation, performance is practically zero when training on 7,000 hours of audio or less, and then follows a roughly log-linear improvement trend till 54,000 hours before also showing diminishing returns when further scaling to the full dataset size.

The general trend across tasks of diminishing returns when moving from 54,000 hours to our full dataset size of 680,000 hours could suggest that the current best Whisper models are under-trained relative to dataset size and performance could be further improved by a combination of longer training and larger models. It could also suggest that we are nearing the end of performance improvements from dataset size scaling for speech recognition. Further analysis is needed to characterize "scaling laws" for speech recognition in order to decided between these explanations.

4.3. Multitask and Multilingual Transfer

A potential concern with jointly training a single model on many tasks and languages is the possibility of negative transfer where interference between the learning of several tasks results in performance worse than would be achieved by training on only a single task or language. To investigate whether this is occurring, we compared the performance of models trained on just English speech recognition with our standard multitask and multilingual training setup and measured their average performance across our suite of zero-shot English speech recognition benchmarks. We adjust for the amount of FLOPs spent training on the task of English speech recognition as only 65% of compute is spent on this task in a joint training setup; analysis would otherwise be confounded by under-training on the task when compared to a same-sized English-only model.

Our results visualized in Figure 9 show that for small models trained with moderate amounts of compute, there is indeed negative transfer between tasks and languages: joint models underperform English-only models trained for the same amount of compute. However, multitask and multilingual